

Writing Small Numbers in Scientific Notation*Question Bank - Additional 3.09*

Show your work. Circle your final answer.

For questions 1 through 4, write each number in scientific notation.

1. 0.008

2. 0.00000026

3. 0.0475

4. 0.000000000091 seconds

For questions 5 and 6, write each number in standard form.

5. 7×10^{-5}

6. 2.6×10^{-8}

Powers of ten.

7. Write each value as a single power of 10, then as a fraction or decimal.

a. 10^{-16} , as a fraction

b. $\left(\frac{1}{10^{-5}}\right)^3$

c. $10^{-9} \cdot 10^6$

8. Two measurements are 7.2×10^{-4} and 0.00000018.

a. Write 7.2×10^{-4} in standard form.

b. Write 0.00000018 in scientific notation.

c. Which of the two is greater? Explain how you know.

9. Complete each comparison.

a. How many times larger is 8.4 than 2.1?

b. How many times larger is 10^{-3} than 10^{-7} ?

c. How many times larger is 8.4×10^{-3} than 2.1×10^{-7} ?

10. Write each value in scientific notation.

a. 112 ten-thousandths

b. 0.00112

c. 112 billionths

11. Which is 0.00067 written in scientific notation?

- ☐ A. 6.7×10^{-4}
- ☐ B. 6.7×10^{-3}
- ☐ C. 67×10^{-5}
- ☐ D. 6.7×10^4

12. Select ALL of the expressions that are equal in value to 0.0034.

- ☐ A. 3.4×10^{-3}
- ☐ B. 34×10^{-4}
- ☐ C. 0.34×10^{-2}
- ☐ D. 340×10^{-5}
- ☐ E. 3.4×10^{-2}
- ☐ F. 3.4×10^3

13. A grain of pollen has a mass of about 8×10^{-9} gram. A bag of flour has a mass of about 2×10^3 grams.

a. How many times larger is the mass of the bag of flour? Write the answer in scientific notation.

b. How many times smaller is the grain of pollen than the bag of flour?